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Sorted on Arrival: Deployment and Ablation of an Agentic Intake Tool for the Campus Lost Property Store

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Abstract. Items handed in at the University's collection points have historically waited on a paper logbook and an infrequent manual review before disposal or donation, a review cycle understaffing left running months behind its own schedule. The Lost Property Ledger replaces the logbook with an agentic intake tool: from a photograph and a short description entered at drop-off, it forecasts an item's reclaim probability, assigns one of five retention tiers, and autonomously schedules a disposal date, all without a staff member drafting a decision. A human-review checkpoint lets staff open any forecast before its scheduled date and amend the tier. We report fourteen months of deployment across six collection points against the fourteen months preceding rollout, and an ablation disabling the review checkpoint at three of the six points throughout the deployment window. Reclaim rates rose in five of six collection points (31% to 36% overall), and the unreclaimed-item backlog, which had nearly quadrupled across the pre-rollout period, declined modestly rather than continuing to grow. Staff opened the review checkpoint for roughly a third of eligible items but amended the forecast tier in very few of those; disabling the checkpoint entirely changed median days-to-reclaim by less than a day (n.s.). We retain the checkpoint for completeness and discuss what its low amendment rate suggests about the work it is actually doing.

Introduction

University lost-property collection — the tray behind the library desk, the tub in the Sports Centre foyer, the shelf in Student Commons — has run on the same instrument for as long as any current staff member can recall: a paper logbook, one line per item, reviewed by whoever has an afternoon free to decide what has sat long enough to donate or discard. That review has never kept pace with intake. Across the six collection points this paper covers, a logbook entry waited a median of several months for its first review, and the backlog of undecided entries grew for fourteen consecutive months before the intervention this paper reports.

The Lost Property Ledger, commissioned by Facilities in response to the backlog, automates the decision the logbook review used to defer. At intake, staff photograph the item and enter a short free-text description; the Ledger forecasts the item's reclaim probability from its description, its collection point, and the point in the teaching calendar it arrived at, sorts it into one of five retention tiers, and schedules a disposal or donation date without further staff input. A review checkpoint remains available: staff may open any pending forecast before its scheduled date and override the tier. Whether that checkpoint does the substantive work its presence implies, or has instead settled into the same deferred-review pattern the Ledger was built to replace, is the question this paper puts to data rather than to intuition. Our contributions:

- we report reclaim-rate and backlog outcomes across fourteen months of Ledger deployment at six collection points, against the fourteen months preceding rollout;
- we describe how often staff actually open the review checkpoint, and how often opening it changes the Ledger's forecast tier;
- we ablate the checkpoint itself, disabling it outright at three of the six collection points

for the full deployment window, and compare days-to-reclaim between the two arms.

Related work

Automation of a previously manual review decision sits inside a well-studied tension between complacency and appropriate reliance: operators of automated decision aids tend toward two failure modes at once, over-trusting a system's output when it performs well and under-trusting it after a single visible error, neither of which tracks the system's actual reliability [1], [2]. Later work qualifies the under-trust finding: people who are given even limited ability to adjust an imperfect algorithm's output continue to rely on it at rates close to constant reliance, suggesting the aversion is partly about control rather than accuracy per se [3] — a finding this paper's review checkpoint, which grants exactly that kind of limited adjustment power, sits close to.

Whether a nominal human-review step changes an automated system's outputs in practice is a separate empirical question from whether reviewers say they value having it. Predictive risk-assessment tools embedded in a human-in-the-loop process have been shown to shift the reviewing human's own judgement toward the tool's recommendation rather than functioning as an independent check on it [4], [5], and a survey of forty-one government policies mandating human oversight of algorithmic decisions found little evidence that the mandated reviewers were positioned, trained, or resourced to meaningfully contest the outputs they were reviewing [6]. The philosophical literature on meaningful human control specifies conditions — a reviewer must track the system's actual reasons, not merely its output — that a checkpoint satisfying only the letter of "a human may intervene" need not meet [7]. Adjacent evidence from clinical alerting systems, where override rates of 49% to 96% have been documented across drug-interaction and dosing alerts, shows that a mandatory human-review step can become habitual rather than deliberative once its volume exceeds what a reviewer can substantively assess [8], [9], [10], [11]. Algorithmic risk-assessment tools more broadly have been found no more accurate than non-expert human judgement on the same task, underscoring that the substantive question is not only whether a human reviews the output but whether either party's judgement is doing better than the other [12]. A systematic review of the empirical literature on perceived fairness of algorithmic decision-making, separately, identifies perceived human involvement in a process as one of the factors most consistently studied for its effect on how fair a decision is judged to be [13]. The governance literature on autonomous, tool-using "agentic" systems — of which the Ledger is a small, low-stakes instance — has begun documenting a comparable oversight gap: an index of deployed agentic systems' technical and safety features tracks, among other properties, whether a system ships a documented mechanism for a human to contest an in-progress action, and a study of developers who hold oversight responsibility over software agents in practice describes the everyday heuristics and challenges of maintaining that oversight as agent throughput scales [14], [15]. On the object side, consumer research on discarded household items finds roughly two-thirds re-enter use through some reuse channel rather than being destroyed, evidence against treating a disposal-date policy as a neutral, valueless step [16].

Method

System. The Ledger runs at the University's six staffed collection points: the Library front desk, the Sports Centre foyer, two lecture-theatre precincts pooled as one point, Student Commons, the Residence Halls office, and the Health Clinic reception. At intake, staff photograph the item and enter a one-line description; the Ledger computes a reclaim-probability estimate $\hat{p}$ from three inputs — a keyword match against the item's description, the collection point, and the current week of the teaching calendar — and assigns the item to one of five retention tiers by thresholding $\hat{p}$:

$$\hat{p}_i = \sigma(\beta_0 + \beta_1 \cdot \mathrm{keyword}_i + \beta_2 \cdot \mathrm{point}_i + \beta_3 \cdot \mathrm{week}_i)$$

Tier assignment fixes a disposal date between two and sixteen weeks out, shorter for low-$\hat{p}$ tiers, without further staff input. The Ledger also autonomously posts a one-line "unclaimed" listing to the Lost & Found noticeboard feed for items above a minimum estimated value, drafted from the same description staff entered — the tool's one genuinely agentic action, in the sense that no draft passes through a person before it is posted.

Review checkpoint. Any pending forecast remains open to staff review until its scheduled disposal date. Opening a forecast surfaces the item's tier, its scheduled date, and the three inputs behind $\hat{p}$; a reviewing staff member may leave the tier unchanged or reassign it to any of the other four, which recomputes the disposal date accordingly.

```
def assign_tier(description, point, week):
    p_hat =
forecast_reclaim_probability(description,
point, week)
    tier = threshold_to_tier(p_hat)
# 5 tiers, lowest = fastest disposal
    disposal_date =
schedule_from_tier(tier)  # 2-16 weeks out
    if tier_value_estimate(description) >=
LISTING_MIN:

post_noticeboard_listing(description,
point)  # autonomous, undrafted by staff
    return tier, disposal_date
```

Ablation. The review checkpoint was disabled outright — forecasts locked against staff amendment — at the Sports Centre, Student Commons, and the Health Clinic for the full fourteen-month deployment window, while the Library, lecture-theatre precinct, and Residence Halls kept the checkpoint open throughout. Collection-point assignment to each arm followed existing staffing rosters rather than random allocation, a design constraint we return to in Limitations.

Windows. We compare fourteen months of logbook-era records preceding rollout against fourteen months of Ledger-era records following it, at the same six collection points, and report days-to-reclaim separately for the checkpoint-enabled and checkpoint-disabled arms across the deployment window.

Results

Reclaim rates rose at five of six collection points. Figure 1 compares reclaim rate by collection point across the two fourteen-month windows. Overall reclaim rate rose from 31% in the logbook era to 36% under the Ledger, concentrated in the Sports Centre (22% to 30%) and Student Commons (35% to 41%); the Health Clinic, the point with the University's smallest item volume, held flat (30% to 31%).

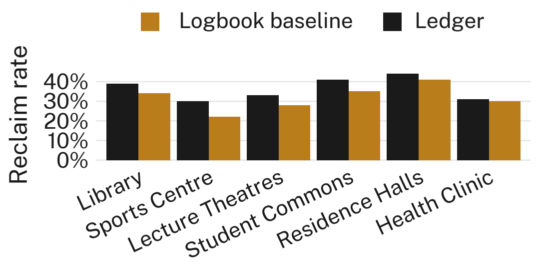

Figure 1: Reclaim rate by collection point, logbook baseline against the fourteen months following the Ledger's rollout: five of six points show a clear rise, one holds flat.

The backlog stopped growing rather than continuing to climb. Figure 2 tracks the unreclaimed-item backlog monthly, from fourteen months before rollout to fourteen months after. The backlog nearly quadrupled across the pre-rollout period, from 50 to 189 items; across the equivalent post-rollout window it declined to 168, an 11% fall against a still-rising rate of new items entering the system.

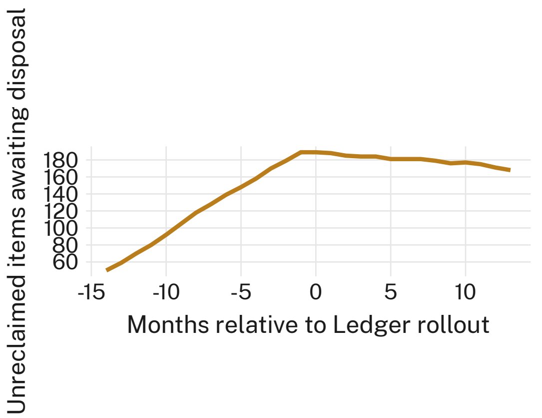

Figure 2: Unreclaimed items awaiting disposal, monthly, relative to the Ledger's rollout at month 0: a fourteen-month climb gives way to a fourteen-month plateau and modest decline.

Staff opened the checkpoint often and amended it rarely. Across the three collection points where the checkpoint remained enabled, staff opened 812 of 2,406 eligible forecasts (34%) before their scheduled disposal date, but amended the forecast tier in only 19 of those 812 openings (2.3% of reviewed items, 0.8% of all eligible items).

The ablation changes almost nothing. Figure 3 compares days-to-reclaim between the checkpoint-enabled and checkpoint-disabled arms across the full deployment window. Median days-to-reclaim was 11 in the enabled arm and 12 in the disabled arm, a difference of well under a day once distributions are compared directly ($p = 0.52$, n.s.); we retain the checkpoint for completeness.

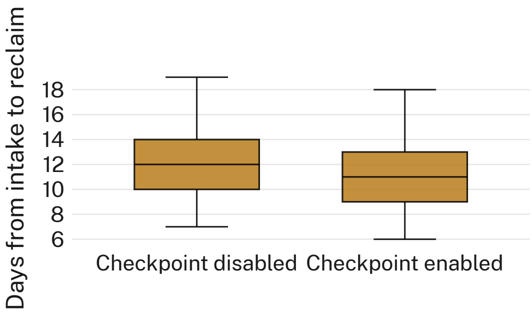

Figure 3: Days from intake to reclaim, collection points with the review checkpoint enabled against the three where it was disabled for the full deployment window: the distributions are effectively indistinguishable.

Discussion

Read together, the three results sit awkwardly against each other in a way we think is itself informative. The Ledger's throughput gains — a flatter backlog, a higher reclaim rate at most points — are consistent with automating a review step that chronic understaffing had left undone rather than done badly [1]. The checkpoint's low amendment rate is consistent with a different, less flattering possibility: that the step functions as the presence of oversight rather than its exercise, a pattern the government-algorithm oversight literature has documented at larger scale and higher stakes than a lost-property store [4], [6]. The ablation result is the sharpest of the three: if disabling the checkpoint changes outcomes negligibly, the checkpoint's 34% open rate was not, on this evidence, doing decision work proportionate to its frequency. We do not read this as evidence the checkpoint should be removed; a low amendment rate is also consistent with the Ledger's forecasts being good enough that little correction is warranted, and our design cannot fully separate the two explanations. Future deployments may benefit from instrumenting what staff actually look at during an open checkpoint, rather than only whether they amend the tier afterward.

Limitations

Collection points were assigned to the ablation's two arms by existing staffing rosters rather than by random allocation, so we cannot rule out that the three checkpoint-disabled points differ from the three enabled points in some way beyond checkpoint status; the two arms' closely matched pre-rollout reclaim rates are some evidence against a large pre-existing difference. Our evaluation covers six collection points over fourteen months at a single university, though the consistency of the throughput gain across five of the six points suggests the pattern is not an artefact of any one location. We did not survey staff about why they open the checkpoint when they do, and so cannot distinguish genuine spot-checking from a habitual click that precedes an unchanged tier in every case.

Conclusion

Fourteen months of Ledger deployment across six collection points shows a flatter unreclaimed-item backlog and a higher reclaim rate at five of six points against the preceding logbook era, alongside a review checkpoint that staff opened for a third of eligible forecasts but amended in under one percent of cases; disabling the checkpoint outright at three collection points changed days-to-reclaim by less than a day. We read the combination as suggesting the checkpoint's value, if any, lies closer to its availability than its exercise. Future work will extend the ablation to the noticeboard listing step, the Ledger's one fully autonomous action, which this paper's design did not separately evaluate.

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